Energy-Efficient Ground Traversability Mapping Based on UAV-UGV Collaborative System

Jianqiang Li, *Member, IEEE*, Yanyan Cheng, Jin Zhou, Jie Chen, Zun Liu, Shuqing Hu, and Victor C. M. Leung, *Life Fellow, IEEE*

***Abstract*—With the development of science and technology, robots have been widely used in smart cities. The traversability mapping of environment perception is the prerequisite for robots to perform tasks. To save the energy consumption of traversability mapping for unmanned ground vehicle (UGV), we fusion a wide range of aerial images and a small amount of ground images to provide vision for UGV. Current map fusion methods are usually constrained by homogeneous model of robotic systems and lack of diverse sensors. As a result, they cannot work well in heterogeneous collaborative robotic systems that consist of aerial and ground robots. In this paper, we use heterogeneous robot systems, including UGV and unmanned aerial vehicles (UAV) to build an occupancy grid map that can be used for navigation. To fuse sensor data of different types, we propose a Collaborative Map Fusion algorithm based on Multi-task Gaussian Process Classification (MTGPC) using heterogeneous robotic systems. Besides, probabilistic model is exploited in traversability mapping, so the active perception can be used to build the map efficiently. Our system is tested in real scenes and can achieve an accuracy of more than 70%. The map fusion using active perception is better than map fusion using random strategy in terms of speed and accuracy. To our knowledge, this is the first work that can build the occupancy grid map using sparse data points sampled from aerial images and ground lidar map.**

***Index Terms*—Map fusion, multi-task Gaussian process classification, heterogeneous robotic system.**

I. Introduction

In MAP-BASED navigation, UGV can quickly determine the current location and navigate to the target location. The demand for autonomous mobile robots such as UGVs and UAVs has greatly increased in recent years, however, it has been proved to be a challenging work for UGVs to navigate in unknown areas due to the uncertainty of the environment [1]. Recently, these robots are deployed to perform various dangerous or repetitive tasks. The robots need information about their surrounding environment to complete the assigned task, which is usually encoded as a traversability map [2]. There are two approaches for the robot to navigate. The first approach assumes that the robot knows nothing about the environment such as the Simultaneous Localization And Mapping (SLAM) [3]–[5] which allows robots to gradually build a complete traversability map while moving in an unknown environment. Generally, exploring an unknown environment will encounter many problems. For example, UGVs could fall into dead ends and need to avoid obstacles, but cameras or lidar detectors cannot detect low-lying lands or stairs in time due to the blind areas. The second approach assumes that the robot performs tasks in a given traversability map [6], this map-based navigation is natural for humans. More specifically, when visiting unknown cities, we only need an existing navigation traversability map in a standard format.

There are many traversability mapping methods in the field of robotics research, and most existing path planning approaches are based on occupancy grid maps which are good choices for ground navigation [7], [8]. Occupancy grids, as they were introduced by Moravec and Elfes(1985), are based upon a regular tessellation of the space into a number of rectangular cells. They store in each cell c of this grid the probability $p(c)$ that the corresponding area in the environment is occupied by an obstacle. The occupancy grid maps include deterministic and probabilistic. The former is to indicate whether there are obstacles at each location, the latter is to indicate the probability of an obstacle at each location and can provide users with better judgment based on probabilistic value. Some studies use node information to construct a topological map for navigation [9]. The topology map contains two basic elements: 1) the convex free space clusters (vertices) and 2) the topological edges connecting adjacent vertices which indicates that a robot can pass between the corresponding vertices. But it is not good at expressing complex structural areas and the shape of obstacles. The 3D texture mapping in Visual Simultaneous Localization And Mapping (VSLAM) is used for UAV applications [10]. It provides a powerful tool for abundant environmental information for navigation and visualization. The precise 3D model allows researchers to interact with the data acquired during the mission and understand the characteristics of the spatial distribution and environmental structure to guide the flight of the UAV. But it cannot be

Manuscript received March 14, 2021; revised July 21, 2021; accepted August 14, 2021. Date of publication August 24, 2021; date of current version February 16, 2022. This work was supported in part by the National Nature Science Foundation of China under Grant U1713212, Grant 62072315, Grant 61836005, Grant 61806130, Grant 62073225, and Grant 62006157; in part by the National Key R&D Program of China under Grant 2020YFA0908700; in part by the Natural Science Foundation of Guangdong Province-Outstanding Youth Program under Grant 2019B151502018; and in part by Public Technology Platform of Shenzhen City under Grant GGFW2018021118145859. *(Corresponding authors: Jie Chen; Victor C. M. Leung.)*

Jianqiang Li, Yanyan Cheng, Jin Zhou, Jie Chen, Zun Liu, and Shuqing Hu are with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China (e-mail: chenjie@szu.edu.cn).

Victor C. M. Leung is with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China, and also with the Department of Electrical and Computer Engineering, University of British Columbia, Vancouver, BC V6T 1Z4, Canada (e-mail: vleung@ieee.org).

Digital Object Identifier 10.1109/TGCN.2021.3107291

used directly for navigation and requires further processing. The data of these traversability maps is a deterministic value, which will cause a problem. When the system does not have enough data to predict the traversability map at a certain place, its predicted value at this place will be extremely uncertain. Therefore, the data is very inaccurate. However, users do not know this situation when using the traversability map but have very high trust in it, which will cause inestimable losses. In this paper, we build a probabilistic traversability map for UGV navigation. They divide the two-dimensional environment space into same-sized grids. Each grid represents a two-dimensional space region in the real environment and the states of grids are occupied or unoccupied.

The world we live in is full of change and unpredictability, so it is difficult to design an autonomous robot that can effectively adapt to all situations. There are many robots of various shapes, sizes and functions, such as UAVs, dual-arm robots [11] and exoskeleton robots [12], [13]. Since different sensors have different advantages and limitations, each robot can carry heterogeneous sensors such as laser scanners, cameras. Although three-dimensional (3-D) lidars have a field of view of 360°, the quality of the maps they generate is not satisfactory due to the sparsity of their physical properties. RGB-D cameras produce denser, more detailed 3-D maps, but their fields of view are limited, and as the sensing range increases, the observation error increases exponentially. Therefore, using homogeneous sensors in multi-robot systems is far from enough in terms of mapping accuracy and robustness [14]. However, employing heterogeneous sensors on multiple robots can make full use of their advantages and provide a more accurate description of the environment. Therefore, multiple robots equipped with different sensors are designed to cooperate with each other to successfully complete complex tasks [15] such as assisting people in environmental monitoring tasks [16], search and rescue tasks [17] and medical care [18]. The air-ground robotic system has the potential to outperform uni-modal robotic systems. We use an air-ground cooperative robot system to combine and take advantage of the aerial and ground robots to generate an occupancy grid map that can be used for navigation of UGVs. The reasons we use an air-ground map fusion are two folds: 1) The lidar maps of the UGVs are accurate, but the observable range of lidar sensors are limited. The UGVs are prone to fall into the low-lying land and dead corners, which leads to low efficiency of traversability mapping [3], [4]. 2) The image maps of UAVs have a wide perspective and low mapping cost, but the accuracy of traversability mapping is low and the two view is easily effected by light or obstacles, such as branches [10]. The proposed method aims to combine the best of two worlds.

The motivation for this paper is that the research of exploiting a multi-robot map fusion framework using Multi-task Gaussian Process(MTGP) has not been studied extensively. In general, the map fusion problem is composed of the estimation of relative transformations and the integration of partial map information, which is usually tackled separately [19]. Existing approaches usually employ point set registration algorithms to achieve map matching [20], [21], where the hard one-to-one correspondence relationship is assigned. Since the intrinsic characteristics (i.e., density. range, field-of-view) of the map vary from different sensors, accurate estimation of the relative transformation is a challenging problem. In this paper, we use a probabilistic model called Multi-task Gaussian Process (MTGP) to solve map fusion of heterogeneous robots (air-ground collaborative robots) and generate an occupied grid map. Firstly, we use the idea of Gaussian process (GP) [22], the predicted value is expressed as a probability value that follows the Gaussian distribution (GD) so that people can be on target when using the traversability map. Then, map fusion using heterogeneous robot teams will dynamically solve complex tasks by effectively combining their complementary functions [23]. However, because the data comes from different perspectives and scopes, how to effectively coordinate heterogeneous teams has been an ongoing challenge in multi-robot research [24], [25]. Multi-task learning is usually used to train prediction models by building correlations between data of different tasks. The occupancy states from air and ground are expressed in discrete binary values, respectively. Based on the data of the UGV and UAV sensors corresponding to the known area, the occupancy probability of the unknown area is predicted. The system can fuse aerial and ground data and can display the results in the form of probability.

The main contributions of this paper are as follows:

1) We propose a Multi-task Gaussian Process Map Fusion (MGPMF) system of heterogeneous robots to fuse aerial and ground data and generate an traversability map. The system can predict occupied probability of unknown areas by establishing the relationship between the image map and the lidar map corresponding to each coordinate in the known areas.
2) We use the MGPMF system based on GP for data fusion to get a probabilistic traversability map. So we can use active perception to guide UGV and UAV to perform path planning and map fusion in the traversability map with high uncertainty, which improves the efficiency of mapping task.
3) Our proposed system has been verified in a real scenario and can achieve an accuracy of more than 70%. The efficiency and accuracy of using active perception mapping are higher than random mapping.

II. Related Work

An accurate traversability map is a prerequisite for the robot to precisely navigate and can ensure that the vehicle can be safely guided without a driver. Among the existing methods of map fusion, some use homogenous data for map fusion, that is, data fusion of multiple UGV sensors or UAV sensors (i.e., multiple cameras) [19], [26]. A global occupancy map is built in real time by incrementally fusing local overlapping Hilbert maps [27]. It uses scans data from multiple UAVs without considering the aerial perspective. Reference [28] consider evidence integration from potentially dependent observation processes under varying spatio-temporal sampling resolutions and noise levels. However, these methods cannot fuse heterogeneous data and take advantage of air-ground collaborative robot systems. The Deep Gaussian Processes(DGP) has a two layer structure that the first layer is responsible for extracting shared and task specific features, while the second layer transforms the features into task specific outputs [29]. It can capture non-linear relationship between the tasks, but the time consumption greatly increased. This paper provides an abstract description of the environment through hand-drawn sketch so that the robot can perform autonomous navigation and exploration when the complete environmental metric description is not available in advance [30]. But it requires a lot of artificial information which is costly. Similarly, a Bayesian framework is proposed that fuses robot motion and visual features for regular 3-D mapping [31]. Besides, a novel idea is proposed for depth estimation from multi-view image-pose pairs [32]. The depth maps can accurately encode the surrounding environmental information, but they cannot be used directly for the navigation of UGVs. The MGPMF system we proposed can fuse air and ground information and generate an occupancy grid map.

Multi-Task Learning (MTL) improves generalization by leveraging the domain-specific information contained in the training signals of related tasks [33]. While the Single-Task GP models the covariance between the values of the same task at different positions, the MTGP allows one to perform simultaneous learning of multiple tasks by modelling the covariance between different tasks at their respective positions [34]. MTGPs perform significantly better than individual GPs at the modeling problem as they effectively integrate heterogeneous sources of information (concentrations of individual elements) to improve the individual predictions of each of them [35]. MTL effectively increases the sample size that we are using for training our model. Regarding air-ground map fusion, the image information captured by the UGVs is easily available. We use MTL for multi-task classification by sharing information between related tasks. The earliest is to use the k-nearest neighbor task clustering algorithm to do multi-task classification [36]. Xue *et al.* derived the distribution from the Dirichlet process and proposed a symmetric multitask learning (SMTL) and asymmetric multitask learning (AMTL) formula to enable the model to learn the similarity between different tasks [37]. A generalized sequential minimal optimization (SMO) algorithm is also used for multi-task classification. Compared with the typical SVM + MTL problem, the proposed SMO can achieve more than 100 times speedup [38]. A deep relationship network is proposed to learn the relationship between tasks [39]. Because the classification category we expect is a probability value, we use the MTGP model to do multi-task classification, which is conducive to active perception in Section III-C.

In addition to air-ground map fusion, another of our improvements is to use active perception to efficiently build maps. A decentralized active sensing algorithm is devised to address the problem of active sensing: How can autonomous robotic vehicles actively cruise a road network to gather and assimilate the most informative data for predicting a spatiotemporally varying traffic phenomenon [40] The sensory system uses a new active phase unwrapping method to effectively determine both 2-D and 3-D sensory information of scenes consisting of multiple objects [41]. However, in existing mapping, Gregorio and Stefano [42] extract elevation maps from laser sensor data and use these maps for navigating an all-terrain vehicle. The elevation map consists of a two-dimensional grid, which stores the height of the surface in the corresponding area in each cell of the discrete grid. Triebel *et al.* [43] introduced multilevel surface maps which is an extension of elevation maps. Compared to elevation maps, multi-level surface maps store in each cell of a discrete grid a list of surfaces and allow mobile robots to operate in environments with multiple levels. Furthermore, Ryde and Hu [44] proposed the volumetric representation with multi-resolution occupied voxel lists. There is an explicit merge or rasterization step to generate a global occupancy map based on the optimized pose [45]. However, as unoccupied areas and unknown areas are not represented, all of the above approaches [42]–[45] cannot update the maps in a probabilistic form. The maps generated by the MGPMF system are probabilistic traversability maps. If the probability of a point is close to 0, the point will be considered unoccupied; if the probability is close to 1, the point will be considered occupied; if the probability is near 0.5, it will be considered an uncertain point.

III. Methodology

The ground traversability mapping proposed can be defined as (1):

$$MP = MGPMF(A_1, A_2, A_3, \ldots, G) \tag{1}$$

where *MP* represents the fusion map of UAVs and UGV, A_i represents the image data of the ith UAV and *G* represents the lidar maps of the UGV, *MGPMF* is the system we propose. The MGPMF system is divided into three parts, image matching, map fusion, and active perception. First, the relationship between the coordinate system of the lidar maps and the image maps is constructed to obtain the initial training data set. Then, the MTGP is used to fuse the heterogeneous image maps and the lidar maps to generate probability occupied grid maps. Finally, based on the generated probabilistic traversability maps, active perception is used to guide UGVs and UAVs for path planning, which improves the efficiency of map fusion. As shown in Fig. 1, the data collection and data processing will be described in detail in Section IV-A.

A. Image Matching

In order to obtain training data, we match the lidar map and the semantic map to obtain the labels for both of them in the same position, as shown in Fig. 1. Image matching needs to consider the scale, translation, and rotation of the two pictures. Regular image matching algorithms include gray-based matching algorithms and feature-based matching algorithms. Perform block matching based on known template images to find similar sub-images in another image. The gray-based matching algorithm is also called the correlation matching algorithm, which uses a two-dimensional spatial sliding template for matching such as Mean Absolute Differences (MAD) [46]. The feature-based matching algorithm first extracts the features of the image, then generates

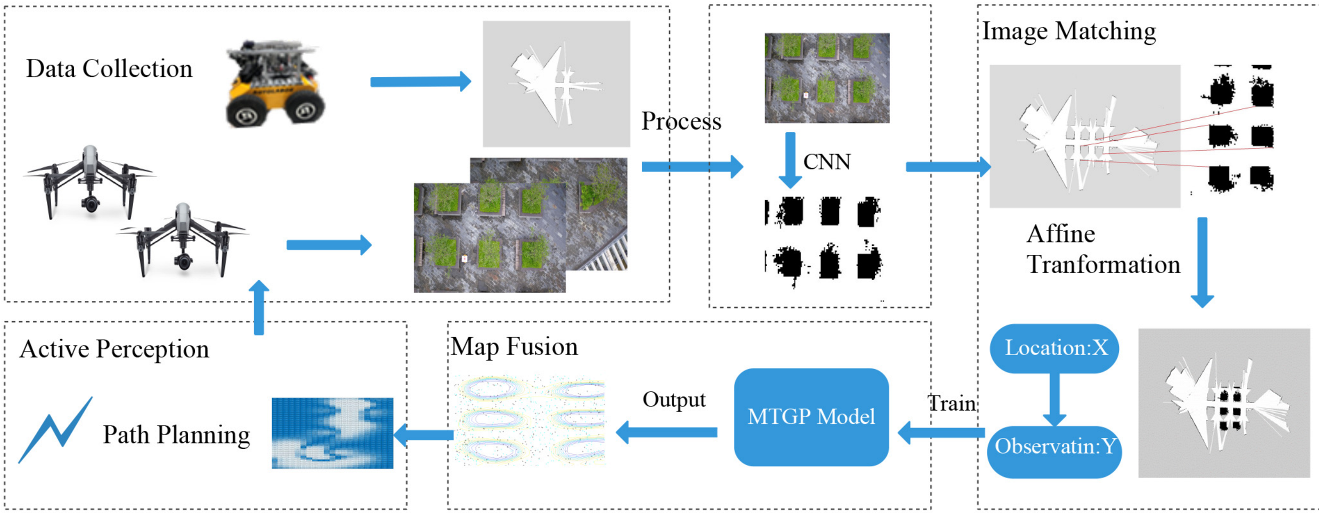


Fig. 1. MGPMF system overview, showing the main steps of map fusion that includes data collection, data processing, model training and prediction.

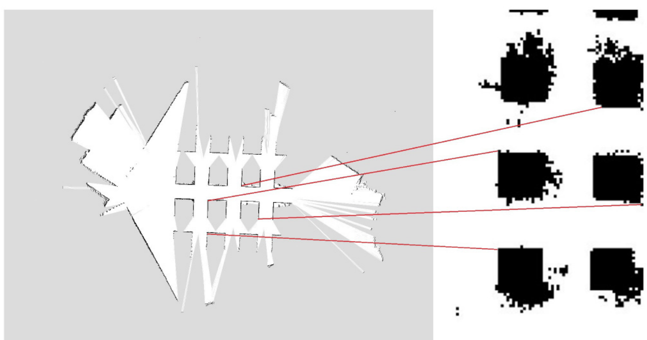


Fig. 2. Manually pairing key points.

a feature descriptor, and finally matches the features of the two images according to the similarity of the descriptors. For example, the Scale-Invariant Feature Transform (SIFT) [47] and Speeded Up Robust Features (SURF) [48]. Unlike traditional image matching algorithms, the images we need to match are two different types, and there are many repeated features (i.e., the flower beds of the same shape), which will cause matching failure. To solve this problem, we use a small amount of manual annotation, that is, calculate the rotation, translation, and scale of the two images through a few matching pairs to complete the image matching, as shown in the Fig. 2. In this experiment, we selected four matching point pairs and can be calculated by

$$\begin{bmatrix} x^a \\ y^a \end{bmatrix} = s \cdot \begin{bmatrix} \cos(a) & -\sin(a) & 0 \\ \sin(a) & \cos(a) & 0 \end{bmatrix} \cdot \begin{bmatrix} x^g \\ y^g \\ 0 \end{bmatrix} + \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \tag{2}$$

Equation (2) is the general form of two-dimensional affine transformation [49], where (x^a, y^a) and (x^g, y^g) are coordinates of two images pixels in a pair of images respectively. The former represents the semantic map, and the latter represents the lidar map. The parameter a represents the rotation angle and the s represents the scaling. Δx and Δy represent the translation values on the x-axis and y-axis between the two maps, respectively. As shown in the Fig. 2, the four pairs of key points are known to the algorithm. Thus the four parameters $(s, a, \Delta x, \Delta y)$ can be solved through formula (8) in [49].

B. Collaborative Map Fusion

In this subsection, the MTGP is used to predict the occupancy probability of the unknown areas by training the labels of the corresponding lidar maps and semantic map at the same position, which can be seen as a binary classification problem. We use the GP to classify the occupied and unoccupied areas. When dealing with data classification problems, it is necessary to establish corresponding classification models for different data sets, that is, traditional Single-Task Learning (STL). However, the STL has the limitation of insufficient information utilization, which leads to a low classification accuracy rate. Since our data set contains four tasks consisting of multiple UAVs and UGVs, Multi-Task Learning (MTL) is used to mine the association between training tasks and perform joint learning on these tasks. Moreover, MTL is an inductive transfer learning that can reduce the number of training samples and training iterations, which is very meaningful for learning tasks with insufficient training samples [33]. In our experiments, each set of tasks contains at least 50 training points. In the classification settings, the target values are discrete. Our data set contains the discrete coordinate points and the corresponding binary values of their image maps and lidar maps. The MTGP binary classification is discussed as follows.

In the general multitask profit classification model, the joint likelihood factorizes as

$$\begin{aligned} p\big(\mathbf{y}, \mathbf{h}, \mathbf{f}, K^t, \alpha | \theta^t, \theta^x\big) = {} & p(\mathbf{y}|\mathbf{h}) p(\mathbf{h}|\mathbf{f}) p\big(\mathbf{f}|K^t, \alpha\big) \\ & p(\alpha|\theta^x) p\big(K^t|\theta^t\big) \end{aligned} \tag{3}$$

$$p(y_i|h_i) = \begin{cases} \delta(h_i)\delta(y_i) & \text{if } y_i = +1 \\ \delta(-h_i)\delta(-y_i) & \text{if } y_i = -1 \end{cases} \tag{4}$$

where $\mathbf{y}$ represents the observations (e.g., $+1$ for unoccupied and -1 for occupied). Because the observations are not available in unvisited areas, we define auxiliary function $\mathbf{h}$ and latent function $\mathbf{f}$ which integrates the information coming from the data and the four tasks (i.e., three UAVs and one UGV) to indirectly predict the observations. Equation (4) represents the relationship between observations $\mathbf{y}$ and the auxiliary variable $\mathbf{h}$, when if the parameter of δ is positive, it is 1, otherwise, it is 0. The auxiliary variable $\mathbf{h}$ conforms to a normal distribution, the mean is $\mathbf{f}$ and the variance is 1. The latent variable $\mathbf{f}$ is represented by a matrix, which conforms to a normal distribution, the mean is 0 and covariance is given by $K^t \otimes K^x$ [50]. The matrix K^t is a task covariance matrix that encodes information about the four tasks, and K^x is a data covariance matrix that encodes information coming from the data of each task (i.e., location coordinate information and binary labels). A task covariance matrix is a free form and the data covariance matrix is given by an automatic relevance determination (ARD) [51] which is a squared exponential covariance with diagonal matrix in this paper. Variables α, θ^t, and θ^x are used to denote prior distributions of parameters of the tasks and data covariance function.

Because the occupancy probability observations $\mathbf{y}$ of the unvisited maps are the unknown values that we want to predict, the non-Gaussian posterior distribution $p(\mathbf{f}|\mathbf{y})$ makes exact inference impossible. In this section, we utilize the EP approximation [52] to do MTGP classification, a deterministic method to approximate the non-Gaussian posterior to solve the above problem.

In this case, we adopt the likelihood given by $p(y|f) = \Phi(yf) = \int_{-\infty}^{yf} \mathcal{N}(0,1)$. Then we can obtain the approximated distribution of the latent variables $q(\mathbf{f}|\mathbf{y})$

$$q(\mathbf{f}|\mathbf{y}) = \frac{1}{Z_{EP}} p(\mathbf{f}) \prod_{i=1}^{n} t_i\left(f_i|\tilde{Z}_i, \tilde{\mu}_i, \tilde{\sigma}_i^2\right) = \mathcal{N}(\mu, \Sigma) \quad (5)$$

where the details are described in [22]. The μ is mean, and the Σ is covariance.

To improve the computational efficiency, the hyperparameters of the tasks and data covariance function in the EP model are estimated by type II maximum likelihood (ML). Moreover, in order to reduce the prediction error, we search for the optimal hyperparameters to minimize the loss. The gradients concerning the hyperparameters θ^x and θ^t are given by

$$\begin{aligned}\frac{\partial \log Z_{EP}}{\partial \theta_j} &= \frac{1}{2}\tilde{\mu}^T\left(K^t \otimes K^x + \tilde{\Sigma}\right)^{-1}\Omega\left(K^t \otimes K^x + \tilde{\Sigma}\right)^{-1}\tilde{\mu} \\ &\quad - \frac{1}{2}\operatorname{tr}\left(\left(K^t \otimes K^x + \tilde{\Sigma}\right)^{-1}\Omega\right) \quad (6)\end{aligned}$$

and

$$\Omega = \begin{cases} K^t \otimes \frac{\partial K^x}{\partial \theta^j} & \text{if } \theta^j = \theta^x \\ \frac{\partial K^t}{\partial \theta^j} \otimes K^x & \text{if } \theta^j = \theta^t. \end{cases} \quad (7)$$

The covariance structure of four tasks is performed through the optimization of the hyperparameters. K^t acts as a task covariance matrix capturing the dependencies between the four tasks. K^x acts as a data covariance matrix capturing the dependencies between the inputs (i.e., the location coordinate information). If we want to restrict the task covariance matrix to be a correlation matrix of the tasks, we have to make sure that the task covariance matrix is symmetric, semi-positive defined. When the off-diagonal elements of K^t are different from zero, the task covariance matrix acts as a transfer of knowledge between tasks. The mean of the predictive distribution of the *j*th task given by

$$\mathbb{E}\left[p\left(y_{j*}|\mathbf{y}\right)\right] = \left(\mathbf{k}_j^t \otimes \mathbf{k}_{\mathbf{x},x_*}^x\right)^T \Sigma^{-1}\mathbf{y} \quad (8)$$

is computed by weighting observations from task *i*, where $i \neq j$, by the *i*th element of $\mathbf{k}_j^t$ [22]. The y_{j*} is the observation to be predicted, that is, the unknown areas that the UGV has not visited are occupied or unoccupied. Variable *i* and *j* are indexes of different tasks. Due to the transfer of knowledge, four tasks are associated. We can use the data in task *i* to predict the observations of task *j*. Therefore, we can use large-scale image maps taken by the UAVs to predict and generate occupied grid maps that the UGV has not visited before.

C. Active Perception

In the above subsection, we propose a Multi-task Gaussian Process Map Fusion (MGPMF) system of heterogeneous robots to generate an occupancy grid map. Because the classification results are expressed in the form of probabilities, we can add an active perception module to improve the experimental results according to the probability of the map occupancy, which will be illustrated in this subsection. We use active learning to calculate the information entropy. The purpose is to eliminate the uncertainty of this event. The higher the information entropy, the higher the uncertainty of this position. Equation (9) is conditional entropy which represents the uncertainty of the label of the unknown region under the condition of the label of the known area. Then we guide the UAV and UGV to do path planning in places with high uncertainty and thus improve the efficiency of prediction by collecting data actively using.

$$H(Y_i|\mathbf{x}_i, L) = -\sum_{y_i} P(y_i|\mathbf{x}_i, L) \log P(y_i|\mathbf{x}_i, L) \quad (9)$$

where function *H* represents the conditional entropy of the unknown label Y_i. $\mathbf{x}_i$ represents the location of observation. *L* is the observed value of this location and y_i is the predicted value of this location. For binary classes, the value of conditional distribution $P(y=1|\mathbf{x}, L)$ is closest to 0.5. We refer to this as the most uncertain.

Fig. 3 represents the entropy map at the current time, the shade of blue represents the size of the entropy value, the darker the blue, the greater the entropy value, the stronger the uncertainty of this point. Because of the occlusion of the trees in the flower bed, the uncertainty around the obstacles will also be very high. When planning the path, the UGV selects a place that can pass and moves forward to a place with a larger entropy value for mapping.

IV. Experiment and Discussion

We have performed a real-world experiment to evaluate the accuracy of our system about map fusion. The performance metric for the prediction accuracy of our experiment is pixel accuracy (PA) which is defined as $PA =$

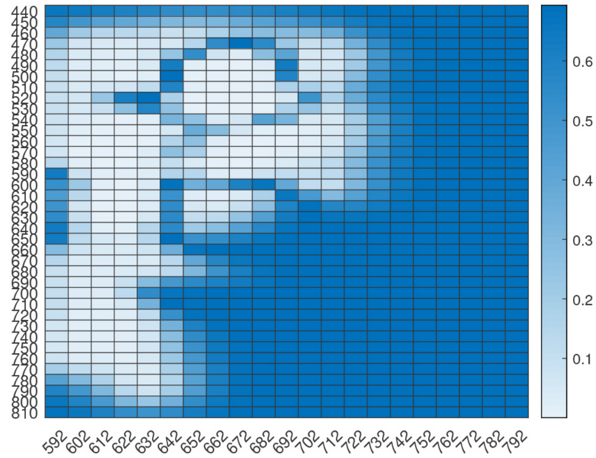


Fig. 3. Visualization of entropy map.

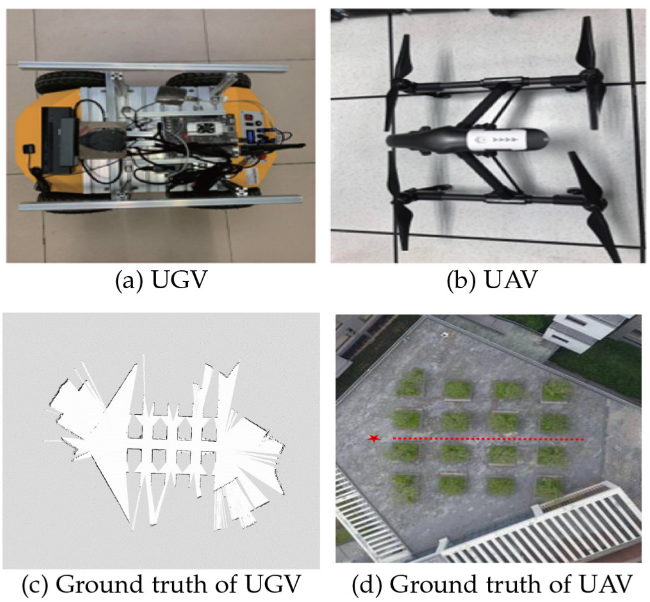


Fig. 4. (a) The UGV of Autolabor Pro 1 and (b) the UAV of DJI Inspire 1 Pro. (c) The ground-truth of UGV and (d) the ground-truth of UAV was collected UAV and UGV respectively.

$\sum_{i=0}^{k} p_{ii} / \sum_{i=0}^{k} \sum_{j=0}^{k} p_{ij}$. That is, the proportion of correctly marked pixels to total pixels. The results are averaged over 10 random instances.

A. Experimental Settings

The MGPMF system is tested in a real scenario with an area of approximately $15 * 15$ square meters, as shown in Fig. 4(d) where the scenario is on the third-floor roof of the School of College of Computer Science and Software Engineering of Shenzhen university and it is captured by the UAV of DJI Inspire 1 Pro in Fig. 4(b). The red star in Fig. 4(d) represents the start position of the UGV, and the red dotted line represents its trajectory. Then we make a SLAM along the red path to obtain the lidar map, which is generated by the UGV of Autolabor Pro 1 in Fig. 4(a) and the sensor onboard the UGV is the rplidar A4, the result is shown in Fig. 4(c). For map fusion, we have collected the image map in Fig. 5(a) taken by the UAV at some point and the lidar map whose resolution is $1344 * 1120$ in pixel in Fig. 5(c) by the SLAM at this point. Besides, the image map in Fig. 5(a) is processed into the semantic graph in Fig. 5(b) through the convolutional neural network (CNN) [53] and the resolution of the semantic map is $372 * 207$ in the grid. As shown in Fig. 5(b), the black area represents the flower bed (i.e., the obstacles area), and the white area represents the concrete floor, that is, the area where the UGV can pass. Note that a blue border is added to distinguish the white area from the background here. The overlapping area in Fig. 5(d) is a fused map of the UAV and the UGV at the same location. The area is matched according to the position of the UGV at this point. Therefore, we obtain the initial training data set.

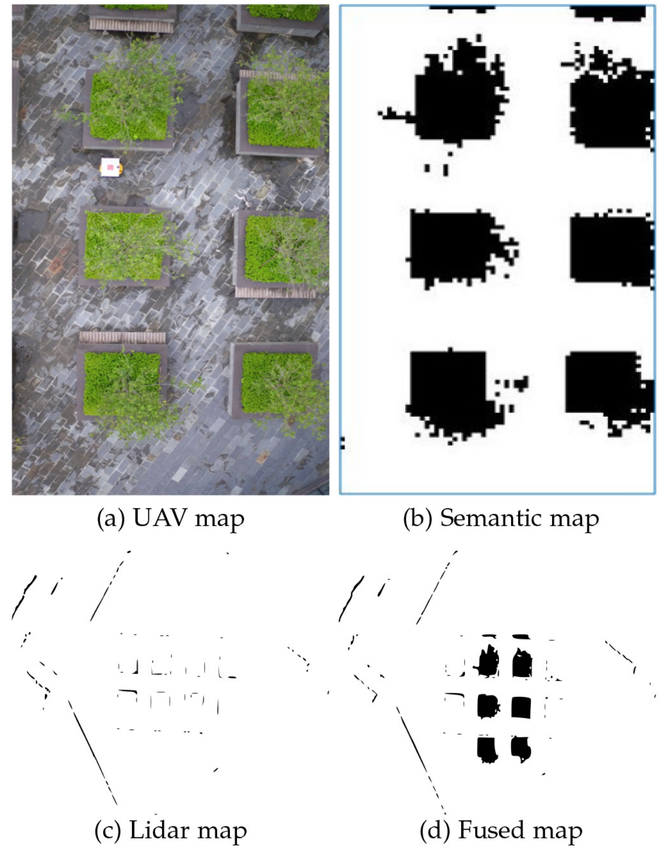


Fig. 5. (a) The image taken by the UAV at some point and (b) The semantic map processed by CNN and (c) the lidar map by SLAM at this point and (d) the fused map.

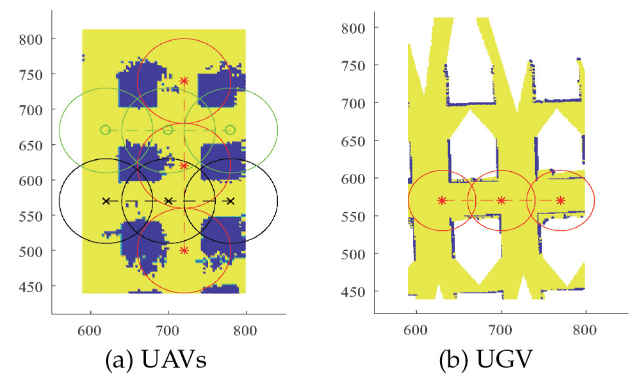


Fig. 6. Visualization of path and radius of perception for UAVs and UGV.

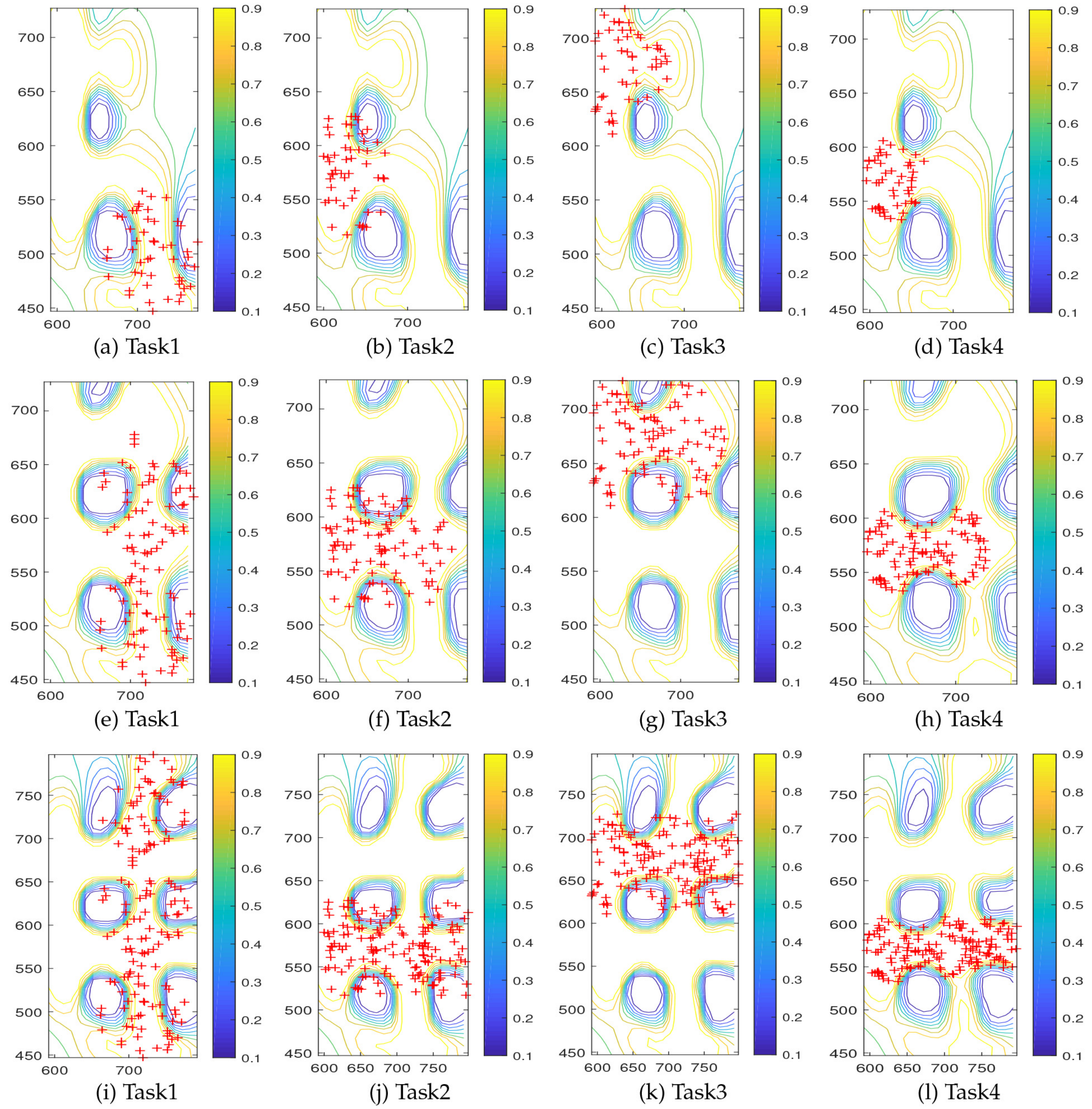


Fig. 7. Prediction probabilities for four tasks (three UAVs and one UGV) in three sets of data respectively.

There are three sets of contract experiment data and each set of data is comprised of four binary classification tasks. The first set of data is collected at the first sampling point by three UAVs and one UGV respectively, including location coordinate information (i.e., GPS) and binary classification labels (i.e., $+1$, -1) corresponding to the location. The second set of data is the sum of the data collected by three UAVs and one UGV at the second sampling point and the first set of data, and so on. That is, in map fusion experiments, one-third, two-thirds, and all training data are trained respectively. Each task contains 50 GPS points and corresponding labels in the first set of data. Each task contains 100 GPS points and corresponding labels in the second set of data and each task contains 150 GPS points and corresponding labels in the third set of data. As shown in Fig. 6, the dashed lines represent the motion trajectories of the robots, the circle areas represent their sampling ranges, and the circled centers represent their sampling points. The red, green, and black colors in Fig. 6(a) represent three different UAVs respectively and their trajectories are from left to right or from bottom to top. Similarly, the UGV moves from left to right in Fig. 6(b).

B. Experimental Results

The fused maps of the MGPMF system in the real scenario are shown in Fig. 7. The red points represent the training data points in each task, and the curves represent the probabilistic values. As demonstrated by the color bar on the right side of each image, the closer the color of curves gets to yellow, the more it likely to be $+1$ (i.e., free area), the closer the

TABLE I
ACCURACY OF PREDICTION RESULTS

Task	Point1 Accuracy	Point2 Accuracy	Point3 Accuracy
task1	0.6153	0.6830	0.7469
task2	0.6163	0.6805	0.7481
task3	0.6153	0.6855	0.7469
task4	0.6153	0.6855	0.7406

TABLE II
THE ACCURACY OF THE THREE SETS OF COMPARATIVE EXPERIMENTS

Method	Point1 Accuracy	Point2 Accuracy	Point3 Accuracy
MMTFL	0.5183	0.5000	0.5183
GO-MTL	**0.6533**	0.4200	0.7233
MGPMF	0.6153	**0.6855**	**0.7406**

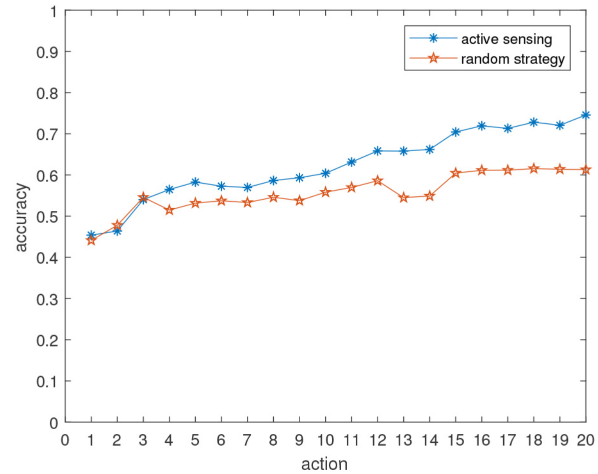


Fig. 8. Comparison of traversability mapping based on active perception and random walk.

color gets to blue, the more likely it to be -1 (i.e., occupied area). If the probability of the curve is close to 0.5, it means that it has the highest uncertainty. Figs. 7(a)–(d) represent the prediction results of the four tasks of the first set of data, Figs. 7(e)–(h) represent the four tasks of the second set of data, and Figs. 7(i)–(l) represent the four tasks of the third set of data. As the number of training points increases, the range of prediction increases. Through the fusion of heterogeneous data, we can use the image maps points of the UAVs and fewer local lidar maps points of the UGV to predict the unknown area that the UGV has not reached.

Table I shows the accuracy of the three sets of comparative experiments. From the table, we can see that as the number of training points increases, the accuracy rate gradually increases. The first three tasks represent three UAVs, and the fourth task represents one UGV. In each set of experiments, the accuracy of four tasks is similar, which indicates that we can predict the unknown area that the UGV has not reached by using UAVs data and a small amount of UGV data. Therefore, our system can help the UGV navigate efficiently and safely in an unknown area.

In Table I, task 1, task2 and task 3 are auxiliary tasks, and the predicted result of task 4 (i.e., UGV) is what we need. So taking the accuracy of task 4 as a reference, we compared the experimental results of the other two methods, which are MMTFL [54] and GO-MTL [55] respectively. The MMTFL decomposes individual task's model parameters into a multiplication of two components. One of the components is used across all tasks and the other component is task-specific. The GO-MTL is a framework for multi-task learning that enables one to selectively share the information across the tasks. As shown in Table II, the optimal results predicted by each set of data are bolded. It can be seen that our method is better than the other two methods. The execution complexity of the proposed method is $O(n^3)$. GO-MTL takes less time and is better than MGPMF when the amount of data is small, but when the amount of data increases, the result of MGPMF is better than it, especially when the amount of data is not very large.

Fig. 8 represents the result of map fusion based on active perception and random strategy. It can be seen from the figure that the active perception strategy can make our system get an occupancy grid map with higher accuracy. In the experimental setup, a UGV and a UAV are used to perform 20 actions, and each action collects 20 points of data respectively. According to the entropy of the probability map generated by the fusion of these points at each action, the strategy plans the next action of the UGV and UAV (set to 30 steps here) to reach the next location and continue to collect data. The system uses the accumulated data fusion to generate a new probability map then iteratively guides the UGV and the UAV to do the next action. As shown in Fig. 8, we analyze the accuracy of twenty actions and compare them with the accuracy of the twenty actions in two strategies. The result is the average of ten experiments.

V. CONCLUSION

To our knowledge, this is the first work that can build the traversability map using sparse data points sampled from aerial images and ground lidar map. This paper proposes a novel MGPMF system. It is the first to fuse a limited amount of heterogeneous data from UAVs and UGV to generate an occupancy grid map in unknown areas that can be used for navigation. The system is tested in a real scenario. As the number of training points increases, the accuracy of the algorithm can reach more than 70%. Then we use the generated probability map to guide the UAV and UGV to perform path planning and map fusion. The map fusion using active perception is better than map fusion using random strategy in terms of speed and accuracy.

REFERENCES

[1] H. Qin *et al.*, "Autonomous exploration and mapping system using heterogeneous UAVs and UGVs in GPS-denied environments," *IEEE Trans. Veh. Technol.*, vol. 68, no. 2, pp. 1339–1350, Feb. 2019.

[2] J. M. Mirats-Tur, C. Zinggerling, and A. Corominas-Murtra, "GIS map based mobile robot navigation in urban environments," in *Proc. IEEE Int. Conf. Adv. Robot.*, 2009, pp. 1–6.

[3] R. Mur-Artal and J. D. Tardós, "ORB-SLAM2: An open-source SLAM system for monocular, stereo, and RGB-D cameras," *IEEE Trans. Robot.*, vol. 33, no. 5, pp. 1255–1262, Oct. 2017.

[4] J. Engel, V. Koltun, and D. Cremers, "Direct sparse odometry," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 3, pp. 611–625, Mar. 2018.

[5] H. Jo, H. M. Cho, S. Jo, and E. Kim, "Efficient grid-based Rao–Blackwellized particle filter SLAM with interparticle map sharing," *IEEE/ASME Trans. Mechatronics*, vol. 23, no. 2, pp. 714–724, Apr. 2018.

[6] S. Cho and S. Hong, "Map based indoor robot navigation and localization using laser range finder," in *Proc. Int. Conf. Control Autom. Robot. Vis.*, 2011, pp. 1559–1564.

[7] D. Meyer-Delius, M. Beinhofer, and W. Burgard, "Occupancy grid models for robot mapping in changing environments," in *Proc. 26th AAAI Conf. Artif. Intell.*, 2012, pp. 2024–2030.

[8] S. Thrun, "Learning occupancy grids with forward models," in *Proc. Int. Conf. Intell. Robots Syst.*, vol. 3, 2001, pp. 1676–1681.

[9] F. Blöchliger, M. Fehr, M. Dymczyk, T. Schneider, and R. Siegwart, "TOPOMAP: Topological mapping and navigation based on visual SLAM maps," 2017. [Online]. Available: arxiv.abs/1709.05533.

[10] X. Li, N. Aouf, and A. Nemra, "3D mapping based VSLAM for UAVs," in *Proc. IEEE Mediterr. Conf. Control Autom.*, 2012, pp. 348–352.

[11] Y. Liu *et al.*, "Motor imagery based teleoperation of a dual-arm robot performing manipulation tasks," *IEEE Trans. Cogn. Develop. Syst.*, vol. 11, no. 3, pp. 414–424, Sep. 2019.

[12] Z. Li *et al.*, "Hybrid brain/muscle signals powered wearable walking exoskeleton enhancing motor ability in climbing stairs activity," *IEEE Trans. Med. Robot. Bionics*, vol. 1, no. 4, pp. 218–227, Nov. 2019.

[13] Z. Li, J. Li, S. Zhao, Y. Yuan, Y. Kang, and C. L. P. Chen, "Adaptive neural control of a kinematically redundant exoskeleton robot using brain–machine interfaces," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 12, pp. 3558–3571, Dec. 2019.

[14] H. Surmann, N. Berninger, and R. Worst, "3D mapping for multi hybrid robot cooperation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2017, pp. 626–633.

[15] Y. Rizk, M. Awad, and E. W. Tunstel, "Cooperative heterogeneous multi-robot systems: A survey," *ACM Comput. Surveys*, vol. 52, no. 2, pp. 1–31, 2019.

[16] F. Shkurti *et al.*, "Multi-domain monitoring of marine environments using a heterogeneous robot team," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2012, pp. 1747–1753.

[17] D. Yanguas-Rojas and E. Mojica-Nava, "Exploration with heterogeneous robots networks for search and rescue," *IFAC PapersOnLine*, vol. 50, no. 1, pp. 7935–7940, 2017.

[18] O. Amir, B. J. Grosz, E. L. M. Law, and R. Stern, "Collaborative health care plan support," in *Proc. ACM Int. Conf. Auton. Agents Multiagent Syst.*, 2013, pp. 793–796.

[19] S. Saeedi, L. Paull, M. Trentini, M. Seto, and H. Li, "Group mapping: A topological approach to map merging for multiple robots," *IEEE Robot. Autom. Mag.*, vol. 21, no. 2, pp. 60–72, Jun. 2014.

[20] Y. Yue, D. Wang, P. Senarathne, and D. Moratuwage, "A hybrid probabilistic and point set registration approach for fusion of 3D occupancy grid maps," in *Proc. IEEE Int. Conf. Syst. Man Cybern. (SMC)*, 2016, pp. 975–980.

[21] J. Jessup, S. N. Givigi, and A. Beaulieu, "Robust and efficient multirobot 3-D mapping merging with octree-based occupancy grids," *IEEE Syst. J.*, vol. 11, no. 3, pp. 1723–1732, Sep. 2017.

[22] G. Skolidis and G. Sanguinetti, "Bayesian multitask classification with Gaussian process priors," *IEEE Trans. Neural Netw.*, vol. 22, no. 12, pp. 2011–2021, Dec. 2011.

[23] E. G. Jones, B. Browning, M. B. Dias, B. Argall, M. Veloso, and A. Stentz, "Dynamically formed heterogeneous robot teams performing tightly-coordinated tasks," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2006, pp. 570–575.

[24] G. A. Kaminka and I. Frenkel, "Towards flexible teamwork in behavior-based robots," in *Proc. 4th Int. Joint Conf. Auton. Agents Multiagent Syst.*, 2005, pp. 1355–1356.

[25] M. Koes *et al.*, "Heterogeneous multirobot coordination with spatial and temporal constraints," in *Proc. AAAI*, vol. 5, 2005, pp. 1292–1297.

[26] D. Zou and P. Tan, "CoSLAM: Collaborative visual SLAM in dynamic environments," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 2, pp. 354–366, Feb. 2013.

[27] K. Doherty, J. Wang, and B. Englot, "Probabilistic map fusion for fast, incremental occupancy mapping with 3D hilbert maps," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2016, pp. 1011–1018.

[28] O. Hamelijnck, T. Damoulas, K. Wang, and M. Girolami, "Multi-resolution multi-task Gaussian processes," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 14025–14035.

[29] A. Boustati, T. Damoulas, and R. S. Savage, "Non-linear multitask learning with deep Gaussian processes," 2019. [Online]. Available: arXiv:1905.12407.

[30] F. Boniardi, A. Valada, W. Burgard, and G. D. Tipaldi, "Autonomous indoor robot navigation using a sketch interface for drawing maps and routes," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2016, pp. 2896–2901.

[31] J. Du, W. Sheng, and M. Liu, "A human–robot collaborative system for robust three-dimensional mapping," *IEEE/ASME Trans. Mechatronics*, vol. 23, no. 5, pp. 2358–2368, Oct. 2018.

[32] Y. Hou, J. Kannala, and A. Solin, "Multi-view stereo by temporal nonparametric fusion," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2019, pp. 2651–2660.

[33] R. Caruana, "Multitask learning," *Mach. Learn.*, vol. 28, no. 1, pp. 41–75, 1997.

[34] S. Vasudevan, "Data fusion with Gaussian processes," *Robot. Auton. Syst.*, vol. 60, no. 12, pp. 1528–1544, 2012.

[35] S. Vasudevan, A. Melkumyan, and S. Scheding, "Information fusion in multi-task Gaussian processes," 2012. [Online]. Available: arXiv:1210.1928.

[36] S. Thrun and J. O'Sullivan, "Discovering structure in multiple learning tasks: The TC algorithm," in *Proc. ICML*, vol. 96, 1996, pp. 489–497.

[37] Y. Xue, X. Liao, L. Carin, and B. Krishnapuram, "Multi-task learning for classification with dirichlet process priors," *J. Mach. Learn. Res.*, vol. 8, no. 1, pp. 35–63, 2007.

[38] F. Cai and V. Cherkassky, "Generalized SMO algorithm for SVM-based multitask learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 23, no. 6, pp. 997–1003, Jun. 2012.

[39] M. Long and J. Wang, "Learning multiple tasks with deep relationship networks," 2015. [Online]. Available: arXiv:1506.02117.

[40] J. Chen, K. H. Low, Y. Yao, and P. Jaillet, "Gaussian process decentralized data fusion and active sensing for spatiotemporal traffic modeling and prediction in mobility-on-demand systems," *IEEE Trans. Autom. Sci. Eng.*, vol. 12, no. 3, pp. 901–921, Jul. 2015.

[41] B. Mobedi and G. Nejat, "3-D active sensing in time-critical urban search and rescue missions," *IEEE/ASME Trans. Mechatronics*, vol. 17, no. 6, pp. 1111–1119, Dec. 2012.

[42] D. De Gregorio and L. Di Stefano, "SKIMAP: An efficient mapping framework for robot navigation," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2017, pp. 2569–2576.

[43] R. Triebel, P. Pfaff, and W. Burgard, "Multi-level surface maps for outdoor terrain mapping and loop closing," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2006, pp. 2276–2282.

[44] J. Ryde and H. Hu, "3D mapping with multi-resolution occupied voxel lists," *Auton. Robots*, vol. 28, no. 2, p. 169, 2010.

[45] N. Fairfield and D. Wettergreen, "Active SLAM and loop prediction with the segmented map using simplified models," in *Proceedings of the Field and Service Robotics*. Berlin, Germany: Springer, 2010, pp. 173–182.

[46] A. A. Nielsen, K. Conradsen, and J. J. Simpson, "Multivariate alteration detection (MAD) and MAF postprocessing in multispectral, bitemporal image data: New approaches to change detection studies," *Remote Sens. Environ.*, vol. 64, no. 1, pp. 1–19, 1998.

[47] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 60, no. 2, pp. 91–110, 2004.

[48] H. Bay, T. Tuytelaars, and L. Van Gool, "SURF: Speeded up robust features," in *Proc. Eur. Conf. Comput. Vis.*, 2006, pp. 404–417.

[49] H. Lin, P. Du, W. Zhao, L. Zhang, and H. Sun, "Image registration based on corner detection and affine transformation," in *Proc. Int. Congr. Image Signal Process.*, vol. 5, 2010, pp. 2184–2188.

[50] E. V. Bonilla, K. M. Chai, and C. Williams, "Multi-task Gaussian process prediction," in *Proc. Adv. Neural Inf. Process. Syst.*, 2008, pp. 153–160.

[51] J. Quinonero-Candela, C. E. Rasmussen, and C. K. Williams, "Approximation methods for Gaussian process regression," in *Large-Scale Kernel Machines*. Cambridge, MA, USA: MIT Press, 2007, pp. 203–223.

[52] M. Opper and O. Winther, "Gaussian processes for classification: Mean-field algorithms," *Neural Comput.*, vol. 12, no. 11, pp. 2655–2684, 2000.

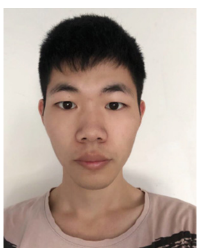
[53] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, 2012, pp. 1097–1105.

[54] X. Wang, J. Bi, S. Yu, J. Sun, and M. Song, "Multiplicative multitask feature learning," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2820–2852, 2016.

[55] A. Kumar and H. Daume, III, "Learning task grouping and overlap in multi-task learning," 2012. [Online]. Available: arXiv:1206.6417.

Jianqiang Li (Member, IEEE) received the B.S. and Ph.D. degrees from South China University of Technology in 2003 and 2008, respectively. He is a Professor with the College of Computer and Software Engineering, Shenzhen University. He led three projects of the National Natural Science Foundation, and three projects of the Natural Science Foundation of Guangdong Province, China. His major research interests include robotic, hybrid systems, Internet of Thing, and embedded systems.

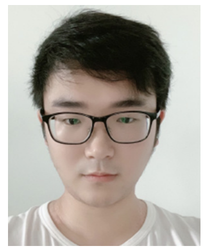
Yanyan Cheng received the B.Eng. degree from Henan University, China, in 2018, and the M.Eng. degree from Shenzhen University, China, in 2021. Her research interests include robotic, machine learning, and heterogeneous collaborative robot mapping.



Jin Zhou received the B.Eng. degree from Jiangsu University, China, in 2020. He is currently pursuing the graduate degree with Shenzhen University. His current research interest is UAV vision slam.

Jie Chen received the B.Sc. degree in EE from Taizhou University, China, in 2006, the M.Sc. degree in CS from Zhejiang University, China, in 2008, and the Ph.D. degree in CS from the National University of Singapore (NUS) in 2013. He joined Shenzhen University as a Assistant Professor in 2021. He was a Postdoctoral Associate with Singapore-MIT Alliance for Research and Technology from 2013 to 2018. He worked with Shenzhen University as a Research-Track Associate Professor from 2018 to 2021. His research interests include machine learning, multiagent system, robotics, intelligent transportation system, and intelligent healthcare system. He received the Dean's Graduate Research Excellence Award from NUS in 2013. He was a Program Committee Member in IJCAI, AAAI, and ICRA.

Zun Liu received the B.S. and Ph.D. degrees from South China University of Technology in 2011 and 2019, respectively. He was a Postdoctoral Fellow with Shenzhen University, China, until 2021, where he is currently a Research-Track Associate Professor. His current research interests include robotic, multiagent system, and reinforcement learning.



Shuqing Hu received the B.Eng. degree from Henan University, China, in 2018, and the M.Eng. degree from Shenzhen University, China, in 2021. His current research direction is multicamera robot Visual SLAM.

Victor C. M. Leung (Life Fellow, IEEE) is a Distinguished Professor of Computer Science and Software Engineering with Shenzhen University, China. He is also an Emeritus Professor of Electrical and Computer Engineering and Director with the Laboratory for Wireless Networks and Mobile Systems, University of British Columbia (UBC), Canada. His research is in the broad areas of wireless networks and mobile systems, and he has published widely in these areas. He received the 1977 APEBC Gold Medal, the 1977–1981 NSERC Postgraduate Scholarships, the IEEE Vancouver Section Centennial Award, the 2011 UBC Killam Research Prize, the 2017 Canadian Award for Telecommunications Research, the 2018 IEEE TCGCC Distinguished Technical Achievement Recognition Award, and the 2018 ACM MSWiM Reginald Fessenden Award. He coauthored papers that won the 2017 IEEE ComSoc Fred W. Ellersick Prize, the 2017 IEEE Systems Journal Best Paper Award, the 2018 IEEE CSIM Best Journal Paper Award, and the 2019 IEEE TCGCC Best Journal Paper Award. He is serving on the Editorial Boards of the IEEE TRANSACTIONS ON GREEN COMMUNICATIONS AND NETWORKING, IEEE TRANSACTIONS ON CLOUD COMPUTING, IEEE TRANSACTIONS ON COMPUTATIONAL SOCIAL SYSTEMS, IEEE ACCESS, *IEEE Network*, and several other journals. He is a Fellow of the Royal Society of Canada (Academy of Science), Canadian Academy of Engineering, and Engineering Institute of Canada. He is named in the current Clarivate Analytics list of "Highly Cited Researchers."